AI 321 Machine Learning for Sequential Data

Fourier Analysis: Continuous and Discrete

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Frequency Domain Analysis

- We have discussed so far about systems in the time domain, with inputs and outputs represented as functions of time
- It is often easier to analyze signals and systems in the *frequency-domain*.
- Analysis of signals and systems primarily consist of:
  - Representing signals as functions of frequency
  - Looking at how systems respond to signals of different frequencies.
  - Developing tools for switching between time-domain and frequency-domain.
  - Learning how to determine which domain is best-suited for a particular problem.

Complex Frequency

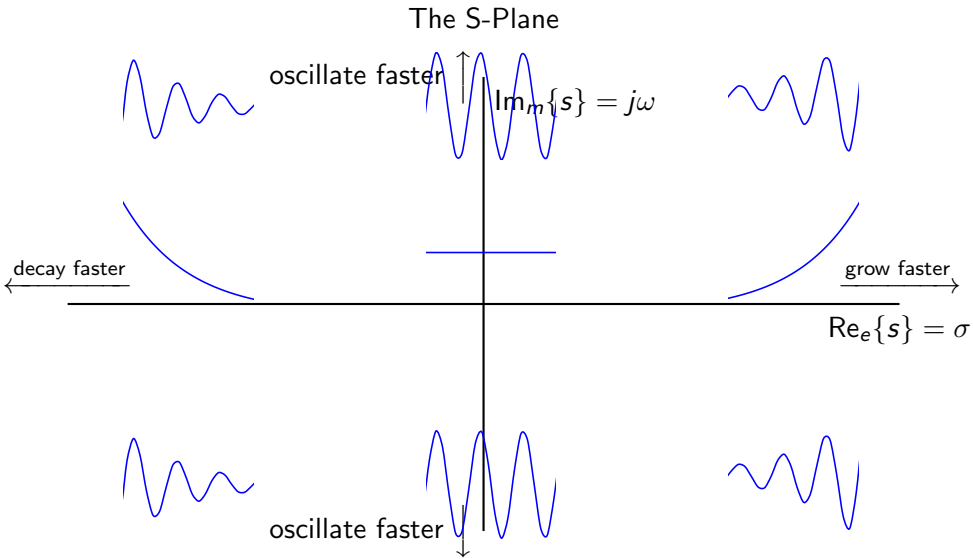
- You learned the concept that "frequency" refers to the number of cycles per second of a periodic signal
- In general, the frequency can be a complex number or "complex frequency".
- Any function written in the form Ke^{st}, where K and s are complex constants, is characterized by the complex frequency "s".
- The complex frequency "s" can be expanded to $s = \sigma + j\omega$

Signals in Complex frequency

Examples of signal described using complex frequency "s". Recall Eulers identity: $e^{j\theta} = cos(\theta) + jsin(\theta)$

Type of Signal	Example	Frequency
DC	$x(t) = 5$	$s = 0$
Exponential	$x(t) = e^{-3t}$	s = -3
Sinusoidal	$x(t) = \sin(50t)$	$s = \pm 50j$
	$= \frac{1}{2j}(e^{50jt} - e^{-50jt})$	
Exponential Sinusoid	$x(t) = e^{2t}\cos(100t)$	$s = 2 \pm 100j$
	$= \frac{1}{2}(e^{(2+100j)t} + e^{(2-100j)t})$	

Signals in Complex frequency



Signals in Complex Frequency

- A sinusoidal or exponential-sinusoidal signal can be expressed as the real part of a single complex exponential as shown:

$$A\cos(\omega t + \theta) = A\,\mathrm{Re}_e\{e^{j\omega t}e^{j\theta}\} = \mathrm{Re}_e\{Be^{j\theta t}\} \text{ (where } B \text{ is complex)}$$

- Using the principles of linearity and superposition, we can find a system's response to a sinusoidal input by finding its response to a corresponding complex exponential input and taking only the real part $\mathrm{Re}_e\{\}$.
- This may seem a lot more work, but functions of the form e^{st}, called *eigenfunctions*, have a special role in CT systems analysis

The Four Fourier Transforms

Table: The Four Fourier Transforms

Fourier Series for Periodic, **Continuous-Time** Signals	Fourier Transform for Aperiodic **Continuous-Time** Signals
Fourier Series for Periodic **Discrete-Time** Signals	Fourier Transform for Aperiodic **Discrete-Time** Signals

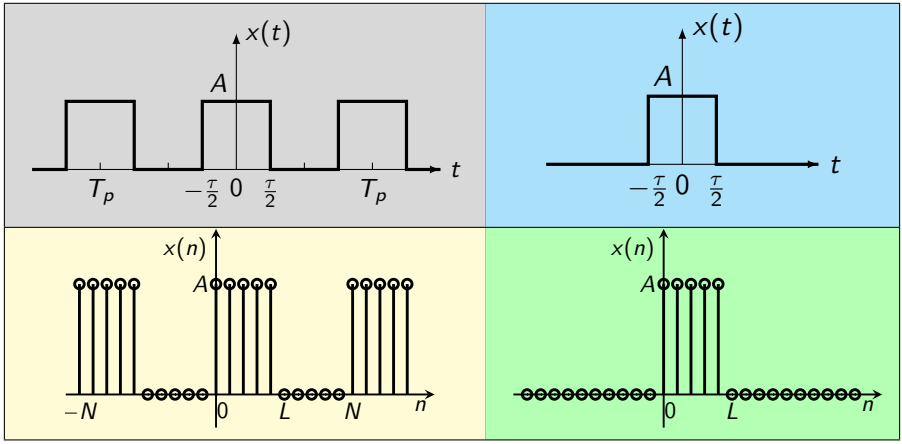
The Four Fourier Transforms

Table: The Four Fourier Transforms

$C_k = \frac{1}{T}\int_{T_0} f(t)e^{-jk\omega_0 t}dt$ $f(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$	$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$ $f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega$
$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N}$ $x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$	$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$ $x(n) = \frac{1}{2\pi}\int_{2\pi} X(\omega)e^{j\omega n}d\omega$

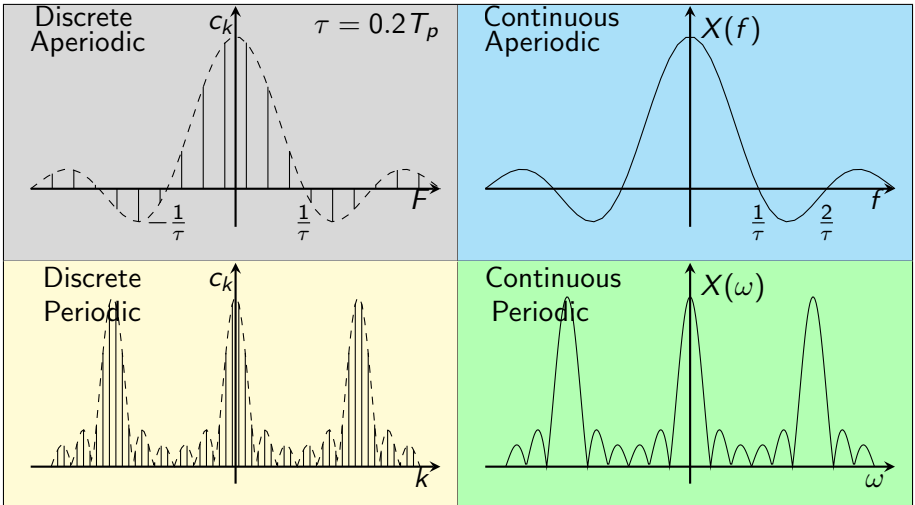
The Four Fourier Transforms

Table: The Four Fourier Transforms

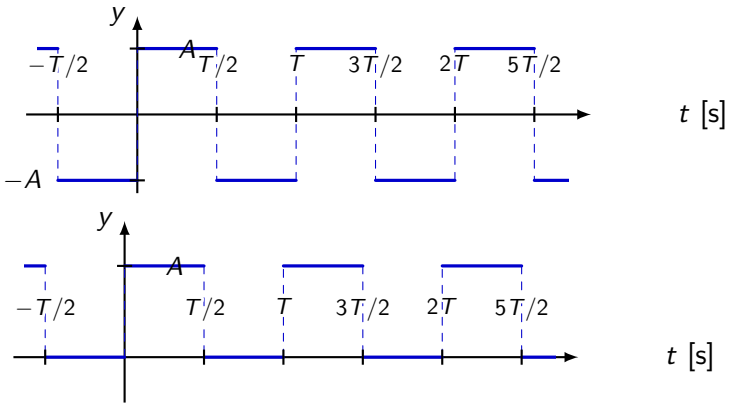


The Spectra of the Four Fourier Transforms

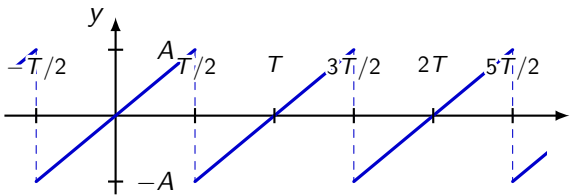
Table: The Four Fourier Transforms

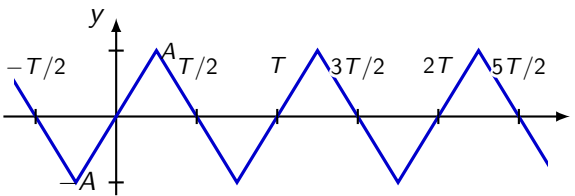


Square Wave and Square Wave with DC Offset

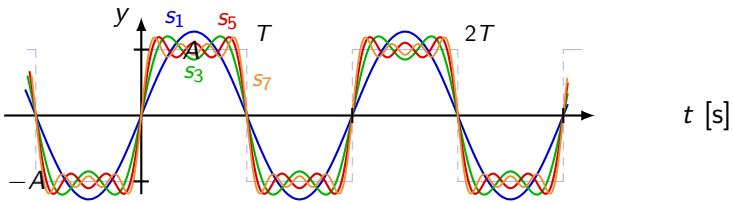


Sawtooth Waveform and Triangular Waveform

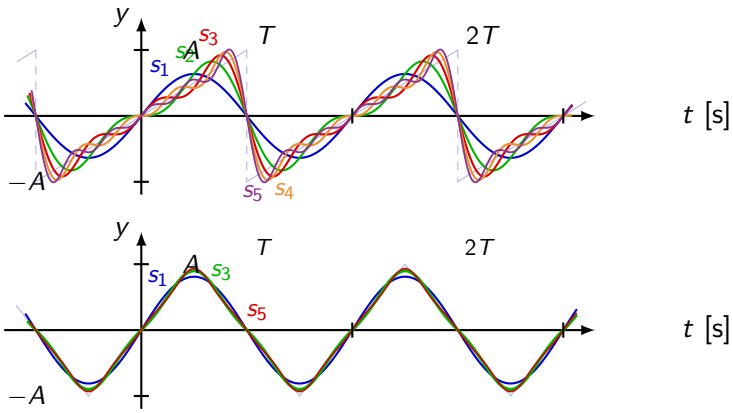




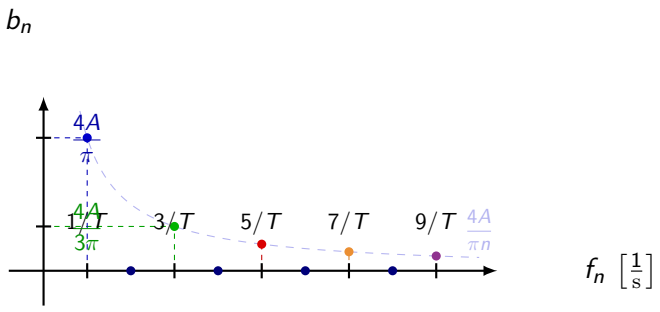
Square Wave Synthesis (time-domain)



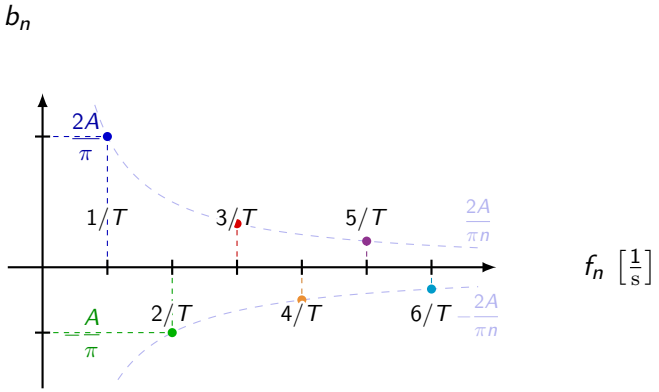
Sawtooth and Triangular Wave Synthesis (time domain)



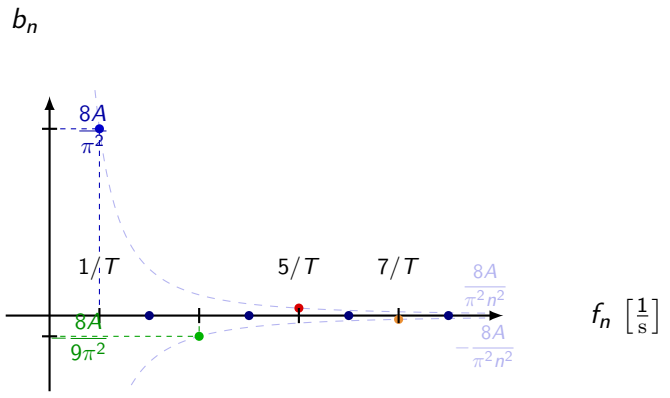
Square Wave Synthesis (frequency domain)



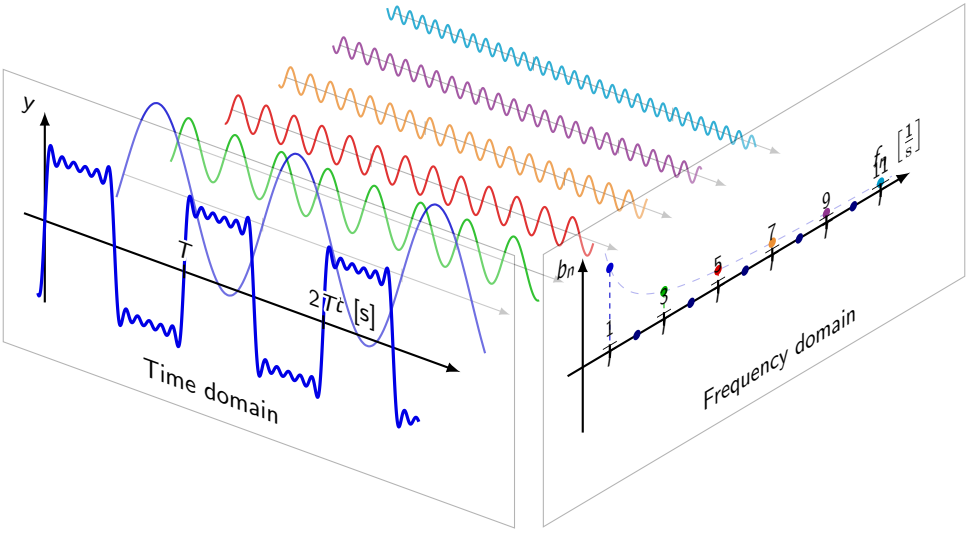
Sawtooth Wave Synthesis (frequency domain)



Triangular Wave Synthesis (frequency domain)



Fourier Synthesis Time and Frequency Domain



Fourier Analysis for DT Systems

- The Fourier Series
- The Fourier Transform

Review of CT Fourier Series

Review: The Fourier series for CT systems, given a periodic signal $x(t)$ with period T.

$$x(t) = \sum_{k=-\infty}^{+\infty} C_k e^{jk\Omega_0 t}$$

where

$$C_k = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jk\Omega_0 t} dt$$

The frequency domain consists of line spectra, representing the magnitude and phase of each C_k Fourier coefficient

Discrete-time Fourier Series

The discrete time Fourier series of a periodic discrete time signal $x(n)$ with period N, that is $x(n+N) = x(n)$, given by:

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \tag{16}$$

The Fourier coefficients are computed as:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \tag{17}$$

Because the signal is discrete in time, the Fourier coefficients are periodic with period N,

$$c_{k+N} = c_k$$

. Eq. (16) is often called the *discrete-time Fourier series (DTFS)*.

Frequency Analysis of Discrete-Time Periodic Signals

Table: Frequency Analysis of Discrete-Time Periodic Signals

Synthesis	$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$	(16)
Analysis	$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$	(17)

Example 1: Discrete-Time Fourier Series

Determine the spectra of the signals.

1. $x(n) = \cos(\sqrt{2}\pi n)$
2. $x(n) = \cos(\pi n/3)$
3. $x(n)$ is periodic with period $N = 4$ and $x(n) = \{\underset{\uparrow}{1}, 1, 0, 0\}$

Solution:

1. $\omega_0 = \sqrt{2}\pi$ and $f_0 = 1/\sqrt{2}$. f_0 is not rational therefore the signal is not periodic, there is no Fourier series representation. Nevertheless, the signal does have a spectra, a single frequency component at $\omega = \omega_0 = \sqrt{2}\pi$

Discrete-Time Fourier Series: Solution (continued)

2 $f_0 = \frac{1}{6}$ therefore periodic with period $N = 6$. From (16), we have

$$c_k = \frac{1}{6}\sum_{n=0}^{5} x(n)e^{-j2\pi kn/6}, \quad k = 0, 1, \ldots, 5.$$

However, $x(n)$ can be expressed as:

$$x(n) = \cos\left(\frac{2\pi n}{6}\right) = \frac{1}{2}e^{j2\pi n/6} + \frac{1}{2}e^{-j2\pi n/6}$$

which is already in the form of an exponential Fourier series, with $c_1 = \frac{1}{2}$ and $c_{-1} = \frac{1}{2}$. The c_{-1} term can be written as:

$$e^{-j2\pi n/6} = e^{j2\pi(5-6)n/6} = e^{j2\pi(5n)/6}$$

where $c_{-1} = c_5$, therefore we have

$$c_0 = c_2 = c_3 = c_4 = 0$$
$$c_1 = \frac{1}{2}, \quad c_5 = \frac{1}{2}$$

Discrete-Time Fourier Series: Solution (continued)

3 From (17), we have

$$c_k = \frac{1}{4}\sum_{n=0}^{3} x(n)e^{-j2\pi kn/4}, \quad k = 0, 1, 2, 3.$$

or

$$c_k = \frac{1}{4}(1 + e^{-j\pi k/2}), \quad k = 0, 1, 2, 3.$$

For $k = 0, 1, 2, 3$ we obtain

$$c_0 = \frac{1}{2}, \quad c_1 = \frac{1}{4}(1 - j), \quad c_2 = 0, \quad c_3 = \frac{1}{4}(1 + j)$$

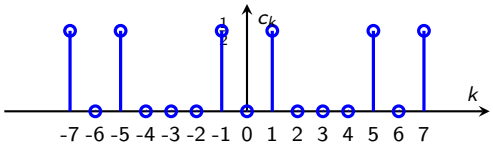
The magnitude and phase spectra are:

$$|c_0| = \frac{1}{2}, \quad |c_1| = \frac{\sqrt{2}}{4} \quad |c_2| = 0, \quad |c_3| = \frac{\sqrt{2}}{4}$$

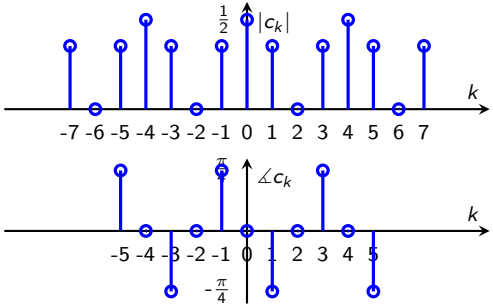
$$\measuredangle c_0 = 0, \quad \measuredangle c_1 = -\frac{\pi}{4}, \quad \measuredangle c_2 = \text{undefined}, \quad \measuredangle c_3 = \frac{\pi}{4}$$

Line Spectra

The line spectra of $\cos(\pi n/3)$ is:

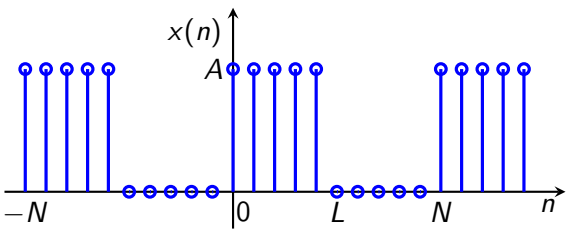


The line spectra for $x(n) = \{1, 1, 0, 0\}$



Example 2: Discrete rectangular periodic function

Determine the Fourier series coefficients and the power spectral density (PSD) of the signal $x(n)$ given below:



Solution

The Fourier series coefficients are computed as:

$$
\begin{aligned}
c_k &= \frac{1}{N}\sum_{n=0}^{N-1} x(n)e^{-j2\pi kn/N} \\
&= \frac{1}{N}\sum_{n=0}^{L-1} Ae^{-j2\pi kn/N} \quad k = 0, 1, \ldots, N-1 \\
&= \frac{A}{N}\sum_{n=0}^{L-1} (e^{-j2\pi k/N})^n
\end{aligned}
$$

which is a **geometric sum**.

Review: Geometric Sum

Recall that if A is the sum of an infinite geometric series,

$$A = \sum_{k=0}^{\infty} a^k = \frac{1}{(1-a)}, \quad \text{if } |a| < 1$$

Proof:

$$A = 1 + a + a^2 + a^3 + \dots \tag{A}$$

$$aA = a + a^2 + a^3 + a^4 + \dots \tag{B}$$

If we subtract equation B from A, we have:

$$(1-a)A = 1, \text{ therefore } A = \frac{1}{(1-a)} \quad |a| < 1$$

Review: Geometric Sum

For a finite duration geometric sequence, we have:

$$A = \sum_{k=0}^{N} a^k = \frac{1 - a^{N+1}}{(1 - a)}$$

Proof:

$$A = 1 + a + a^2 + \ldots + a^N \tag{C}$$

$$aA = a + a^2 + A^3 + \ldots + a^{N+1} \tag{D}$$

Again, we subtract equation D from C, we have

$$(1 - a)A = 1 - a^{N+1}, \text{ therefore } A = \frac{1 - a^{N+1}}{(1 - a)}$$

Solution

From the sum of a finite duration geometric series, we have:

$$c_k = \frac{A}{N}\sum_{n=0}^{L-1}(e^{-j2\pi k/N})^n = \begin{cases} \dfrac{AL}{N}, & k = 0 \\ \dfrac{A}{N}\dfrac{1-e^{-j2\pi kL/N}}{1-e^{-j2\pi k/N}}, & k = 1, 2, \ldots, N-1 \end{cases}$$

The last expression can be simplified further to:

$$\begin{aligned} \frac{A}{N}\frac{1-e^{j2\pi kL/N}}{1-e^{j2\pi k/N}} &= \frac{A}{N}\frac{e^{-j\pi kL/N}}{e^{-j\pi k/N}}\frac{e^{j\pi kL/N}-e^{-j\pi kL/N}}{e^{j\pi k/N}-e^{-j\pi k/N}} \\ &= \frac{A}{N}e^{-j\pi k(L-1)/N}\frac{\sin(\pi kL/N)}{\sin(\pi k/N)} \end{aligned}$$

Power Spectral Density (PSD)

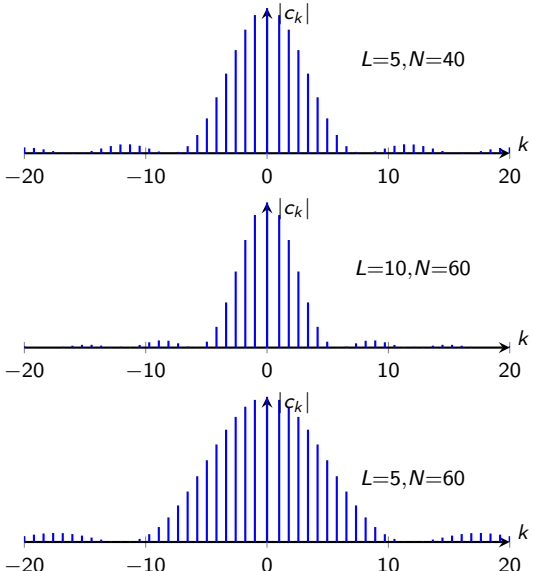
The average power contained in a discrete-time periodic signal is:

$$P_\xi = \sum_{k=0}^{N-1} |c_k|^2 = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

Therefore, we have:

$$|c_k|^2 = \begin{cases} \left(\frac{AL}{N}\right)^2, & k = 0, \pm 1, \pm 2, \ldots, \\ \left(\frac{A}{N}\right)^2 \left(\frac{\sin(\pi k L/N)}{\sin(\pi k/N)}\right)^2, & \text{otherwise} \end{cases}$$

PSD of Rectangular Pulse (A=1)



Insights

- As L increases. the width of the main lobe of the line spectra decreases
- As N increases the number of samples in the line spectra increases

Example 3: DT Fourier Series

Given $x(n) = \cos\left(\frac{2\pi}{3}n\right) + \sin\left(\frac{2\pi}{5}n\right)$, find c_k.

Solution: The common period of sine and cosine terms is $N = 15$, then using Euler's identity

$$x(n) = \left(\frac{e^{j\frac{2\pi}{3}n} + e^{-j\frac{2\pi}{3}n}}{2}\right) + \left(\frac{e^{j\frac{2\pi}{5}n} - e^{-j\frac{2\pi}{5}n}}{j2}\right)$$

$$= \left(\frac{\overbrace{e^{j2\pi\frac{5}{15}n}}^{k=5} + \overbrace{e^{j2\pi\frac{10}{15}n}}^{k=10}}{2}\right) + \left(\frac{\overbrace{e^{j2\pi\frac{3}{15}n}}^{k=3} - \overbrace{e^{j2\pi\frac{12}{15}n}}^{k=12}}{j2}\right)$$

Which is already in the form of an exponential sum, with the indicated values of k.

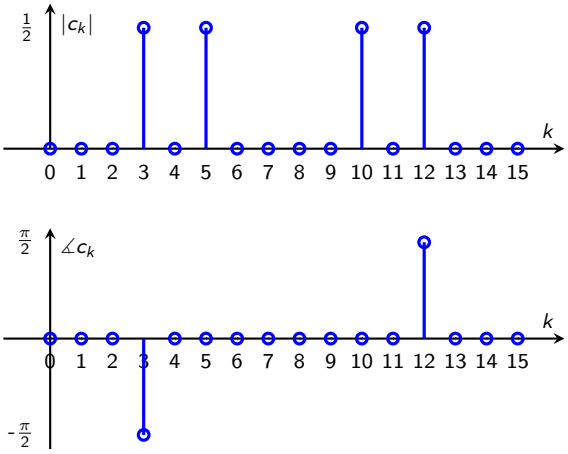
Example 3: DT Fourier Series (solution)

Therefore,

$$c_k = \begin{cases} \frac{1}{2}, & k = 5, 10 \\ \frac{1}{j2}, & k = 3 \\ \frac{-1}{j2}, & k = 12 \\ 0, & \text{otherwise} \end{cases}$$

Note that $c_{10} = c_{-5}$ and $c_{12} = c_{-3}$

Example 3: DT Fourier Series (solution)



Example 4: (Optional) Homework

Given $x(n) = \cos\left(\frac{2\pi}{3}n\right)\sin\left(\frac{2\pi}{5}n\right)$, find c_k.

You can work out the solution.

Answer:

$$c_k = \begin{cases} \frac{-1}{j4}, & k = 2, 7 \\ \frac{1}{j4}, & k = 8, 13 \\ 0, & \text{otherwise} \end{cases}$$

Discrete-Time Fourier Transform

The Fourier transform of a finite-energy discrete time signal is defined as:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \tag{18}$$

- $X(\omega)$ represents the frequency content of the signal $x(n)$, and is a continuous function of ω.
- $X(\omega)$ is a decomposition of $x(n)$ into its frequency components
- $X(\omega)$ has a range of $(-\pi, \pi)$ or $(0, 2\pi)$ and is periodic:

$$X(\omega + 2\pi k) = X(\omega)$$

Discrete-Time Inverse Fourier Transform

The inverse Fourier Transform is given as:

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \tag{19}$$

Frequency Analysis of Discrete Time Aperiodic Signals

Table: Frequency Analysis of Discrete-Time Aperiodic Signals

Synthesis	$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$	(19)
Analysis	$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$	(18)

Uniform Convergences of the Fourier Transform

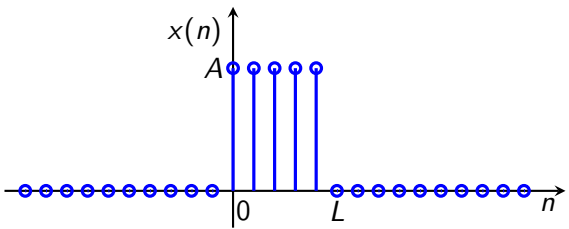
Uniform convergence of the DTFT is guaranteed if $x(n)$ is absolutely summable, that is:

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

Example 1: Aperiodic rectangular pulse

Determine the Fourier transform and the energy density spectrum of the sequence:

$$x(n) = \begin{cases} A, & 0 \le n \le L-1 \\ 0, & \text{otherwise} \end{cases}$$



Example 1; Solution

1. Check if $x(n)$ is absolutely summable.

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{L-1} |A| = L|A| < \infty$$

2. The Fourier transform of $x(n)$ exists and can be computed as:

$$X(\omega) = \sum_{n=0}^{L-1} Ae^{-j\omega n}$$

3. At $\omega = 0$,

$$X(\omega) = \sum_{n=0}^{L-1} Ae^{0} = AL$$

Example 1: Solution

4. At $\omega \neq 0$,

$$\begin{aligned} X(\omega) &= A\frac{1-e^{-j\omega L}}{1-e^{-j\omega}} \\ &= Ae^{-j(\omega/2)(L-1)}\frac{\sin(\omega L/2)}{\sin(\omega/2)} \end{aligned}$$

5. The magnitude of $X(\omega)$ is

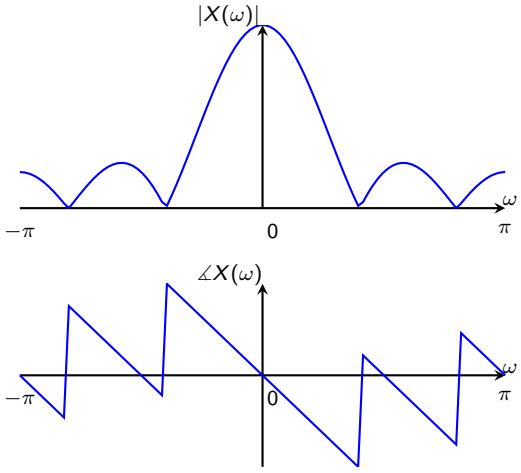
$$|X(\omega)| = \begin{cases} |A|L, & \omega = 0, \\ |A|\left|\dfrac{\sin(\omega L/2)}{\sin(\omega/2)}\right|, & \text{otherwise.} \end{cases}$$

and the phase is:

$$\measuredangle X(\omega) = \measuredangle A - \frac{\omega}{2}(L-1) + \measuredangle\frac{\sin(\omega L/2)}{\sin(\omega/2)}$$

Magnitude and Phase of $X(\omega)$

If $x(n)$ is real, the magnitude exhibits **even** symmetry and the phase exhibits **odd** symmetry.



Parseval's relation for DT Aperiodic signals

The energy of a finite-duration aperiodic signals is:

$$\begin{aligned} E_x &= \sum_{n=-\infty}^{\infty} |x(n)|^2 \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 \, d\omega \end{aligned}$$

The **energy density spectrum** is given by:

$$S_{xx}(\omega) = |X(\omega)|^2$$

Symmetric Properties of the Fourier Transform

Given that $x(n)$ has a Fourier transform $X(\omega)$, and that they can be both complex, can be expressed in rectangular form as real and imaginary parts:

$$X(\omega) = X_R(\omega) + X_I(\omega)$$

$$x(n) = x_R(n) + x_I(n)$$

The following relationships hold:

$$X_R(\omega) = \sum_{n=-\infty}^{\infty} \{x_R(n)\cos(\omega_n) + x_I(n)\sin(\omega_n)\}$$

$$X_I(\omega) = -\sum_{n=-\infty}^{\infty} \{x_R(n)\sin(\omega_n) - x_I(n)\cos(\omega_n)\}$$

Symmetric Properties of the Fourier Transform

The following relationships hold for $x(n)$:

$$x_R(n) = \frac{1}{2\pi} \int_{2\pi} [X_R(\omega)\cos(\omega_n) - X_I(\omega)\sin(\omega_n) d\omega]$$

$$x_I(n) = \frac{1}{2\pi} \int_{2\pi} [X_R(\omega)\sin(\omega_n) + X_I(\omega)\cos(\omega_n) d\omega]$$

Hermitian symmetry for real signals:

$$X^*(\omega) = X(-\omega)$$

Properties of the Fourier Transform

Given the following Fourier Transform pairs:

$$x(n) \overset{\mathcal{F}}{\Longleftrightarrow} X(\omega)$$

$$x_1(n) \overset{\mathcal{F}}{\Longleftrightarrow} X_1(\omega)$$

$$x_2(n) \overset{\mathcal{F}}{\Longleftrightarrow} X_2(\omega)$$

The following properties hold:

1. Linearity property

$$a_1 X_1(\omega) + a_2 X_2(\omega) \overset{\mathcal{F}}{\Longleftrightarrow} a_1 x_1(n) + a_2 x_2(n)$$

Properties of the Fourier Transform

2. Time shifting

$$e^{-j\omega k}X(\omega) \overset{\mathcal{F}}{\Longleftrightarrow} x(n-k)$$

3. Time reversal

$$X(-\omega) \overset{\mathcal{F}}{\Longleftrightarrow} x(-n)$$

4. Convolution Theorem

$$X_1(\omega)X_2(\omega) \overset{\mathcal{F}}{\Longleftrightarrow} x_1(n)^* x_2(n)$$

Properties of the Fourier Transform

5. Correlation Theorem.
   The correlation of $x_1(n)$ and $x_2(n)$ is:

$$r_{x_1x_2}(n) = \sum_{k=-\infty}^{\infty} x_1(k)x_2(k-n)$$

$$r_{x_1x_2}(n) \overset{\mathcal{F}}{\Longleftrightarrow} S_{x_1x_2}(\omega) = X_1(\omega)X_2(-\omega)$$

6. Wiener-Khintchine Theorem
   Let $x(n)$ be a real signal

$$r_{xx}(l) \overset{\mathcal{F}}{\Longleftrightarrow} S_{xx}(\omega)$$

   The energy spectral density is the Fourier transform of the autocorrelation sequence.

Properties of the Fourier Transform

7. Frequency shifting

$$X(\omega - \omega_0) \overset{\mathcal{F}}{\Longleftrightarrow} e^{j\omega_0 n} x(n)$$

8. Modulation Theorem

$$\frac{1}{2}\left[X(\omega + \omega_0) + X(\omega - \omega_0)\right] \overset{\mathcal{F}}{\Longleftrightarrow} x(n)\cos(\omega_0 n)$$

Properties of the Fourier Transform

9. Parseval's Theorem

$$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\omega)X_2^*(\omega)d\omega$$

10. Windowing Theorem

$$x_3(n) \equiv x_1(n)x_2(n) \overset{\mathcal{F}}{\Longleftrightarrow} X_3(\omega) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega-\lambda)d\lambda$$

11. Differentiation in the frequency domain

$$j\frac{dX(\omega)}{d\omega} \overset{\mathcal{F}}{\Longleftrightarrow} nx(n)$$

Example 2

Given $x(n) = \{\underset{\uparrow}{-1}, 2, 3\}$ and $h(n) = \{\underset{\uparrow}{1}, 1\}$.

Convolve $x(n)$ and $h(n)$ to get $y(n)$.

$$x(n) * h(0) = \{-1, 2, 3\}$$
$$x(n) * h(1) = \{0, -1, 2, 3\}$$

$$y(n) = x(n) * h(n) = x(n) * h(0) + x(n) * h(1) = \{-1, 1, 5, 3\}$$

Example 3

Frequency domain: $X(\omega) = -1 + 2e^{-j\omega} + 3e^{-j2\omega}$ and $H(\omega) = 1 + e^{-j\omega}$

$$\begin{aligned} Y(\omega) &= X(\omega)H(\omega) \\ &= -1 + 2e^{-j\omega} + 3e^{-j2\omega} - e^{-j\omega} + 2e^{-j2\omega} + 3e^{-j3\omega} \\ &= -1 + e^{-j\omega} + 5e^{-j2\omega} + 3e^{-j3\omega} \end{aligned}$$

$$y(n) = \mathcal{F}^{-1}\{Y(\omega)\} = \{-1, 1, 5, 3\}$$

Example 4:

Determine the energy density spectrum $S_{xx}(\omega)$ of the signal:

$$x(n) = a^n u(n) \quad -1 < a < 1$$

Solution:

1. Since $|a| < 1$, the sequence us absolutely summable, hence the Fourier transform exists.

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{\infty} |a|^n = \frac{1}{1-|a|} < \infty$$

2. The Fourier transform exists and is given by:

$$X(\omega) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

Since $|ae^{-j\omega n}| = |a| < 1$, we have

$$X(\omega) = \frac{1}{1-ae^{-j\omega}}$$

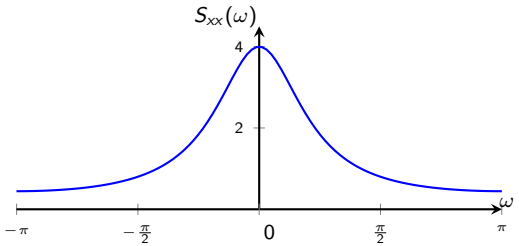
Example 4

3. The density spectrum is given by

$$S_{xx}(\omega) = |X(\omega)|^2 = X(\omega)X(-\omega) = \frac{1}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}$$

$$S_{xx}(\omega) = \frac{1}{1 - 2a\cos(\omega) + a^2}$$

4. Plot of $S_{xx}(\omega)$ with $a = 0.5$.



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